This absorption causes excitation of d-electrons of central metal ion from the lower energy t_{2g} level to the higher energy e_g level which is known as d-d transition.

Let us understand the d-d transitions by considering $[Ti(H_2O)_6]^{3+}$ as an example. In this complex the central metal ion is Ti^{3+}, which has d^1 configuration. This single electron occupies one of the t_{2g} orbitals in the octahedral aqua ligand field. When white light falls on this complex the d electron absorbs light and promotes itself to e_g level. The spectral data show the absorption maximum is at 20000 cm^{-1} corresponding to the crystal field splitting energy (Δ_o) 239.7 kJ mol^{-1}. The transmitted colour associated with this absorption is purple and hence ,the complex appears purple in colour.

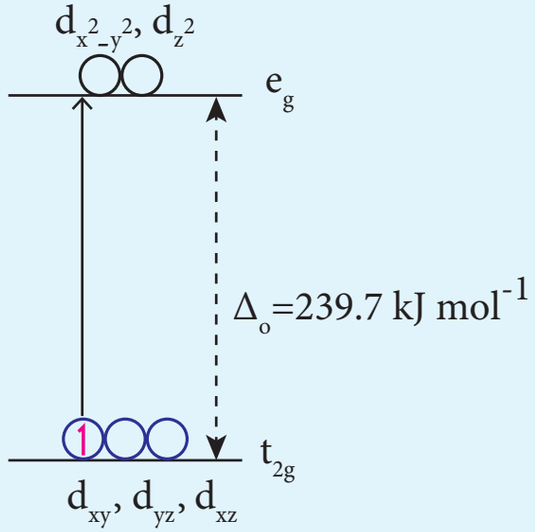


Figure 5.16 d-d Transition

The octahedral titanium(III) complexes with other ligands such as bromide and fluoride have different colours. This is due to the difference in the magnitude of crystal field splitting by these ligands (Refer page 156). However, the complexes of central metal atom such as of Sc^{3+}, Ti^{4+}, Cu^{+}, Zn^{2+}, etc... are colourless. This is because the d-d transition is not possible in complexes with central metal having d^0 or d^{10} configuration.

Evaluate yourself 8:

i. The mean pairing energy and octahedral field splitting energy of $[Mn(CN)_6]^{3-}$ are 28,800 cm^{-1} and 38500 cm^{-1} respectively. Whether this complex is stable in low spin or high spin?

ii. Draw energy level diagram and indicate the number of electrons in each level for the complex $[Cu(H_2O)_6]^{2+}$. Whether the complex is paramagnetic or diamagnetic?

iii. For the $[CoF_6]^{3-}$ ion the mean pairing energy is found to be 21000 cm^{-1} . The magnitude of Δ_0 is 13000cm^{-1}. Calculate the crystal field stabilization energy for this complex ion corresponding to low spin and high spin states.

Metallic carbonyls

Metal carbonyls are the transition metal complexes of carbon monoxide, containing Metal-Carbon bond. In these complexes CO molecule acts as a neutral ligand. The first homoleptic carbonyl $\left[Ni(CO)_4\right]$ nickel tetra carbonyl was reported by Mond in 1890.These metallic carbonyls are widely studied because of their industrial importance, catalytic properties and their ability to release carbon monoxide.

XII U5 Coordination jagan.indd 160 22-12-2022 19:51:06